

Engineering healthier dreams

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Dreaming is abnormal in nearly every psychological and neurodevelopmental condition, from PTSD to depression to autism. Yet there is no scientific consensus on why we dream. This impasse is due largely to challenges historically facing dream researchers, such as the difficulty of uncovering evidence to convincingly nail down why we dream. It seems we will forever remain in the dark about the adaptive functions of dreaming.

What if experiments could be run to systematically influence dream content in real time? **Dream engineering** refers to a set of techniques used to influence dreaming. By engineering dreams, we can now study their functions more rigorously and perhaps also find solutions for when dreams go awry.

One method to engineer dreams involves surreptitiously presenting stimuli that are incorporated into ongoing dreams. For instance, spraying water on a participant's hand may increase their likelihood of dreaming about water. When car noises reach sleepers without waking them up, they may be more likely to dream of traffic.

Sounds played during sleep can be strategically selected. When a sound was previously linked with a specific memory, the sound can reactivate that memory during sleep. Dreaming aside, this method, called **targeted memory reactivation**, has been used in more than 100 recent experiments to impact memory processing. The method is also promising for engineering dreams.

In one study conducted at the University of Montreal, participants enjoyed flying in a virtual reality game before sleep. Sounds from that experience were then presented during sleep. As a result, they dreamed more about the flying task over the next week and improved their game playing.

In a study by our team at Northwestern University, participants engaged in two unusual activities in the lab before sleep, one involving blowing bubbles and the other playing a harmonica. Then, during REM sleep, the sound associated with one activity was presented, and it increased dreaming about that specific activity. Those dreams were also linked with increased creativity and decreased negative emotion pertaining to that activity.

We took it a step further in another experiment by recruiting **lucid dreamers** — individuals who become aware of dreaming while remaining in the dream. Before sleep, they worked on brain-teaser puzzles, each linked with a specific sound, until they failed to solve

four puzzles. While they slept, we presented sounds associated with two of the unsolved puzzles. We told them that if they heard a puzzle's sound in their dream, they should try to solve it, such as by exploring the puzzle scenario in their dream. Dreams incorporated cued puzzles, and 42% of those dreamed puzzles were solved, compared with 17% of puzzles that were not incorporated into dreams.

The study also used another type of dream engineering, **two-way communication**. When dreamers heard puzzle cues within their dreams, they let us know by completing rapid in-out sniffs. We monitored sniffs by measuring airflow from their nostrils, so they were able to inform us about their ongoing dream while remaining asleep. Researchers in Paris have also shown that sleepers can communicate outside of lucid dreams, such as by twitching facial muscles to correctly categorize words and pseudowords, for example, “pizza” vs. “ditza.”

Another method for engineering dreams is **incubating dreams**, or focusing on a topic before sleep with the intention to dream about it. In one experiment involving naps, when people were repeatedly reminded to think of a specific topic while they fell asleep, their dream reports revealed that over 90% experienced a dream related to the target topic. People feel empowered when they realize that they can have an active role in their own dreaming, which can help combat nightmares. Furthermore, this method can easily be applied in the home with minimal technology. More generally, there may be great value in spending the few minutes before going to sleep wisely, given that what's on our minds during this critical time may seed our dreams throughout the night, whether we wake up remembering them or not.

In sum, engineering dreams may be a new entry point for optimizing sleep and mental health. It may also help us gain a better understanding of why we dream at all. 🌈

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